

ABDULLA ALASMAWI

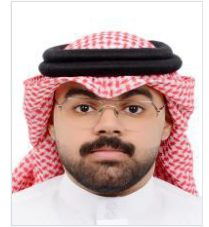
Software Engineer · Full-Stack & Cloud

B.Sc. Computer Science (Cloud Computing), University of Bahrain · Class of 2026

[Go] [TypeScript] [React] [Python] [AWS]

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// PROFILE

Software engineer who builds full-stack products and runs them on AWS: the interface, the API, the data layer and the deployment, mainly in Go, TypeScript and Python. Most recently built the Ministry of Interior Police K9 Unit's monitoring platform at the AWS Cloud Innovation Center, and a bilingual registry piloted across four Shura Council teams. Cares about shipping software real teams use, backed by tests and reviewed code. Looking to join a product team and own features from design to production.

// PROFESSIONAL EXPERIENCE

Network & Information Security Intern · Bahrain Shura Council

Jul 2026 – Present

- ▶ Built Bayyan, the department's bilingual English and Arabic register of its telecom, permit, contract and subscription obligations: a NestJS and Prisma API, a React front end and PostgreSQL, deployed on Linux behind nginx (see Projects).
- ▶ Moved a production MariaDB database off Amazon RDS to on-premises servers through an S3 snapshot, and sequenced the cutover so the restored copy matched the source with no data lost.
- ▶ Traced a kernel version incompatibility that stopped the AWS Elastic Disaster Recovery agent from replicating a production EC2 instance from Ireland to Frankfurt, and worked the case with AWS Support until the agent installed.
- ▶ Support staff with network, endpoint and infrastructure issues; run account lifecycle across Active Directory, Entra ID and Intune; and roll out PAM, BitLocker encryption policy and ManageEngine patching across the fleet.
- ▶ Built a site presenting the organization's HR achievements for a GCC-level awards competition, published on GitHub Pages.
- ▶ Write the technical documentation and step-by-step procedures the department works from after handover.

Cloud & IoT Intern · AWS Cloud Innovation Center (CIC)

Feb 2026 – Jun 2026

- ▶ Built the Ministry of Interior Police K9 Unit's monitoring platform end to end: the dashboard, the serverless backend, and the ingestion pipeline between them.
- ▶ Designed the backend on AWS: IoT Core and API Gateway take the data in, Lambda processes it, DynamoDB stores it, and Cognito keeps handlers and kennel staff to their own views.
- ▶ Wired in Amazon Bedrock to flag anomalies in the sensor data and turn them into care recommendations the dashboard shows as plain guidance.
- ▶ Demonstrated the finished system live to AWS mentors and Ministry of Interior stakeholders.

// SELECTED PROJECTS

[K9 Pavlov System](#) ↗ · Senior Capstone, built out at AWS CIC

- ▶ One place to see each police dog's health, the conditions in its kennel and who is assigned to it, replacing paper records and separate checks.
- ▶ A custom ESP32 collar, kennel sensors and a smart food scale report through a Raspberry Pi 5 gateway to AWS IoT Core over MQTT, where the CIC backend above takes over.

[AWS IoT Core] [Lambda] [DynamoDB] [Amazon Bedrock] [Cognito] [ESP32] [React]

Bayyan · Services & Obligations Registry (Shura Council)

- ▶ Internal web app that gives the Networks & Information Security department one register of its telecom, permit, contract and subscription obligations, piloted across Network Operations, Information Security, Infrastructure and the Service Desk.
- ▶ Each record's billing cycle drives its renewal date, and records turn to warning and then critical as that date approaches, so nothing lapses unnoticed.
- ▶ Fully bilingual with a right-to-left Arabic layout, document attachments in MinIO, and PDF and Excel reports through WeasyPrint and ExcelJS; runs natively on Linux behind nginx and systemd.

[NestJS] [Prisma] [React] [PostgreSQL] [MinIO] [nginx]

Guidely ↗ · Internal Knowledge Assistant

- ▶ Answers everyday questions about a team's own documents in a sentence instead of twelve pages of policy, and shows the exact passages each answer came from so the reader can check it.
- ▶ Documents are split along their headings, embedded and searched with FAISS, and answered fully locally through sentence-transformers and Ollama or through the OpenAI API. Finds the right passage for 94% of test questions; 264 automated tests.

[Python] [FastAPI] [FAISS] [sentence-transformers] [Ollama] [SQLite] [React]

Brain-Book ↗ · Full-Stack Social Network

- ▶ A social platform where people post, join groups, share photos and message each other instantly. Built the React front end, the Go backend and the REST API between them, with WebSockets for live delivery and SQLite behind sign-in and privacy settings.
- ▶ Built and defended in a live audit at Reboot01, where the code is reviewed by peers before it passes.

[Go] [React] [WebSockets] [REST APIs] [SQLite]

Detecto ↗ · Person Detection & Counting

- ▶ Counts the people in an image, a video or a live webcam feed with a YOLOv8 model behind a FastAPI service, and keeps every run for review with occupancy statistics, zone alerts, and CSV and Excel export.

[Python] [FastAPI] [YOLOv8] [OpenCV] [React] [SQLite]

- ▶ **Also built from scratch:** an HTTP/1.1 server on epoll, a Unix shell and a ray tracer in Rust, a networked multiplayer FPS, and a JavaScript framework with a virtual DOM — all at github.com/Alasmawi and alasmawi.dev.

// TECHNICAL SKILLS

LANGUAGES	Go, TypeScript, JavaScript, Python, Rust, Java, C++
FRONTEND	React, Vite, Tailwind CSS, Framer Motion, right-to-left layouts, responsive design
BACKEND	NestJS, Prisma, FastAPI, REST APIs, WebSockets
CLOUD (AWS)	Lambda, API Gateway, IoT Core, DynamoDB, Cognito, S3, RDS, EC2, Elastic Disaster Recovery
DATA & AI	PostgreSQL, MariaDB, MySQL, SQLite, FAISS, sentence-transformers, Ollama, YOLOv8, Amazon Bedrock
DEVOPS & TOOLS	Linux, Docker, nginx, systemd, MinIO, Git, automated testing, peer code review
IT & SECURITY	Active Directory, Entra ID, Intune, PAM, BitLocker, ManageEngine
SPOKEN	Arabic (native), English (fluent)

// EDUCATION

B.Sc. Computer Science, Cloud Computing · University of Bahrain 2022 – 2026

- ▶ Specialized in cloud architecture, security, networking and distributed systems, on a track built around the AWS Academy curriculum. Senior capstone: the K9 Pavlov System, carried into the AWS Cloud Innovation Center internship.
- ▶ Coursework: Cloud Computing Architecture, Cloud Development, Cloud Security, Internet Software Development, Database Management Systems, Analysis and Design of Algorithms, Parallel and Distributed Computing, Operating Systems.

Full Stack Development · Reboot Coding Institute W.L.L 2024 – 2026

- ▶ Project-based on the O1-edu model, studied alongside the degree: Go, algorithms, networking and full-stack web development in the Common Core, now in the DevOps specialization.
- ▶ No lectures or written exams: every project is peer-reviewed and defended in a live audit before it passes. Brain-Book was built and audited here.